Comparative Analysis of Chatbot Architectures with and without Memory

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Motivation

BUSINESS • DEALS

Nebius, Meta Agree to $27 Billion AI Infrastructure Pact

Nebius said it will provide $12 billion of dedicated capacity across multiple locations

THE WALL STREET JOURNAL.

TECHNOLOGY • ARTIFICIAL INTELLIGENCE • KEYWORDS: CHRISTOPHER MIMS

The Global Memory-Chip Shortage Will Cost Us All

AI companies' need for a type of once-affordable microchip threatens to drive up prices of all electronics—and limit data-center ambitions

The Electric Grid Needs Huge Upgrades. No One Knows Who Will Pay for Them.

Utilities around the U.S. are set to spend tens of billions of dollars on high-voltage lines, largely to meet demand from data centers

Motivation

What is Long-Term Memory (in this study)?
Long-term memory refers to a chatbot's ability to persistently store and recall information from past interactions across sessions, enabling context retention and personalization over time.



TECHNOLOGY

A Memory-Chip Shortage Is Squeezing Consumer Tech—and It's Set to Get Worse

Higher component costs mean bigger price tags, smaller profit margins and less memory in some consumer electronics

The Problem

- Memory is widely assumed to improve chatbot performance
- Significant resources are invested in memory systems

However:

- Limited empirical evidence exists
- Impact varies across task types

Approach / Methodology Overview

Compare two chatbot architectures:

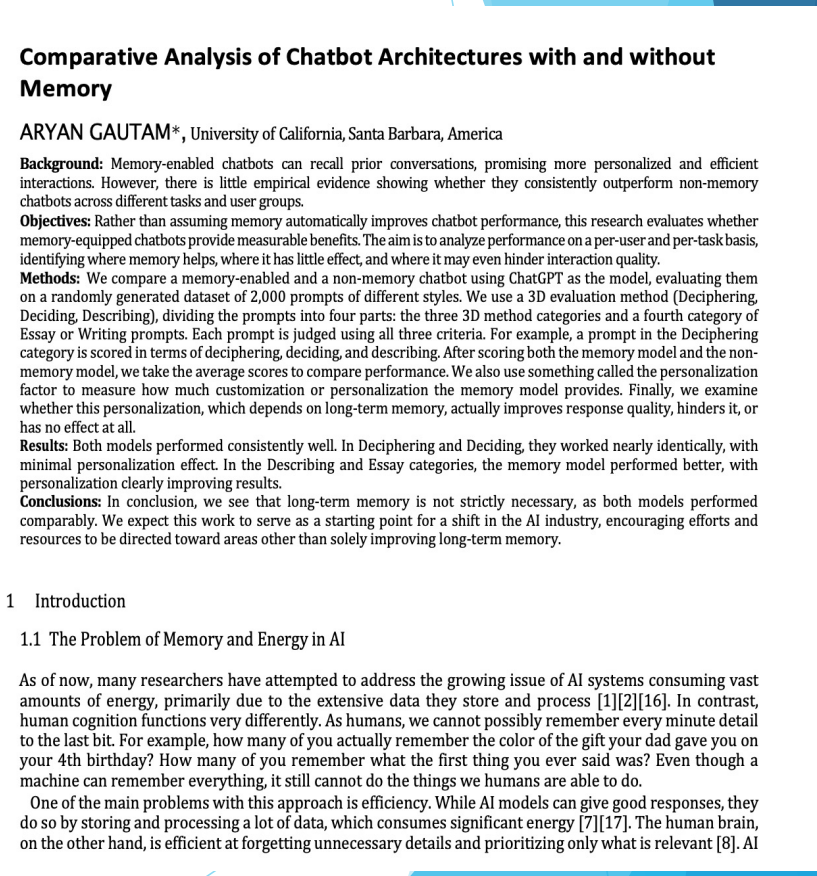
- Memory-enabled model
- Non-memory model

Evaluate across:

- 2,000 diverse prompts

Use structured evaluation:

- 3D Framework
- Personalization Factor



Comparative Analysis of Chatbot Architectures with and without Memory

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Background: Memory-enabled chatbots can recall prior conversations, promising more personalized and efficient interactions. However, there is little empirical evidence showing whether they consistently outperform non-memory chatbots across different tasks and user groups.
Objectives: Rather than assuming memory automatically improves chatbot performance, this research evaluates whether memory-equipped chatbots provide measurable benefits. The aim is to analyze performance on a per-user and per-task basis, identifying where memory helps, where it has little effect, and where it may even hinder interaction quality.
Methods: We compare a memory-enabled and a non-memory chatbot using ChatGPT as the model, evaluating them on a randomly generated dataset of 2,000 prompts of different styles. We use a 3D evaluation method (Deciphering, Deciding, Describing), dividing the prompts into four parts: the three 3D method categories and a fourth category of Essay or Writing prompts. Each prompt is judged using all three criteria. For example, a prompt in the Deciphering category is scored in terms of deciphering, deciding, and describing. After scoring both the memory model and the non-memory model, we take the average scores to compare performance. We also use something called the personalization factor to measure how much customization or personalization the memory model provides. Finally, we examine whether this personalization, which depends on long-term memory, actually improves response quality, hinders it, or has no effect at all.
Results: Both models performed consistently well. In Deciphering and Deciding, they worked nearly identically, with minimal personalization effect. In the Describing and Essay categories, the memory model performed better, with personalization clearly improving results.
Conclusions: In conclusion, we see that long-term memory is not strictly necessary, as both models performed comparably. We expect this work to serve as a starting point for a shift in the AI industry, encouraging efforts and resources to be directed toward areas other than solely improving long-term memory.

1 Introduction

1.1 The Problem of Memory and Energy in AI

As of now, many researchers have attempted to address the growing issue of AI systems consuming vast amounts of energy, primarily due to the extensive data they store and process [1][2][16]. In contrast, human cognition functions very differently. As humans, we cannot possibly remember every minute detail to the last bit. For example, how many of you actually remember the color of the gift your dad gave you on your 4th birthday? How many of you remember what the first thing you ever said was? Even though a machine can remember everything, it still cannot do the things we humans are able to do.

One of the main problems with this approach is efficiency. While AI models can give good responses, they do so by storing and processing a lot of data, which consumes significant energy [7][17]. The human brain, on the other hand, is efficient at forgetting unnecessary details and prioritizing only what is relevant [8]. AI

Evaluation Framework (3D Framework)

We evaluate chatbot performance using three dimensions:

- **Deciphering**
  Understanding and extracting information
- **Deciding**
  Reasoning and selecting the correct outcome
- **Describing**
  Explaining concepts clearly

Additional category:

- Essay / Writing tasks

We evaluate chatbot performance using three dimensions:

- **Deciphering**

"The baker decorated the cake. What did the baker decorate?"

- **Deciding**

"What happens to a candle flame when covered with a glass jar?"

- **Describing**

"What is machine learning?"

Additional category:

- Essay / Writing tasks

"Write an essay on the effects of advertising on consumer behavior."

Deciphering

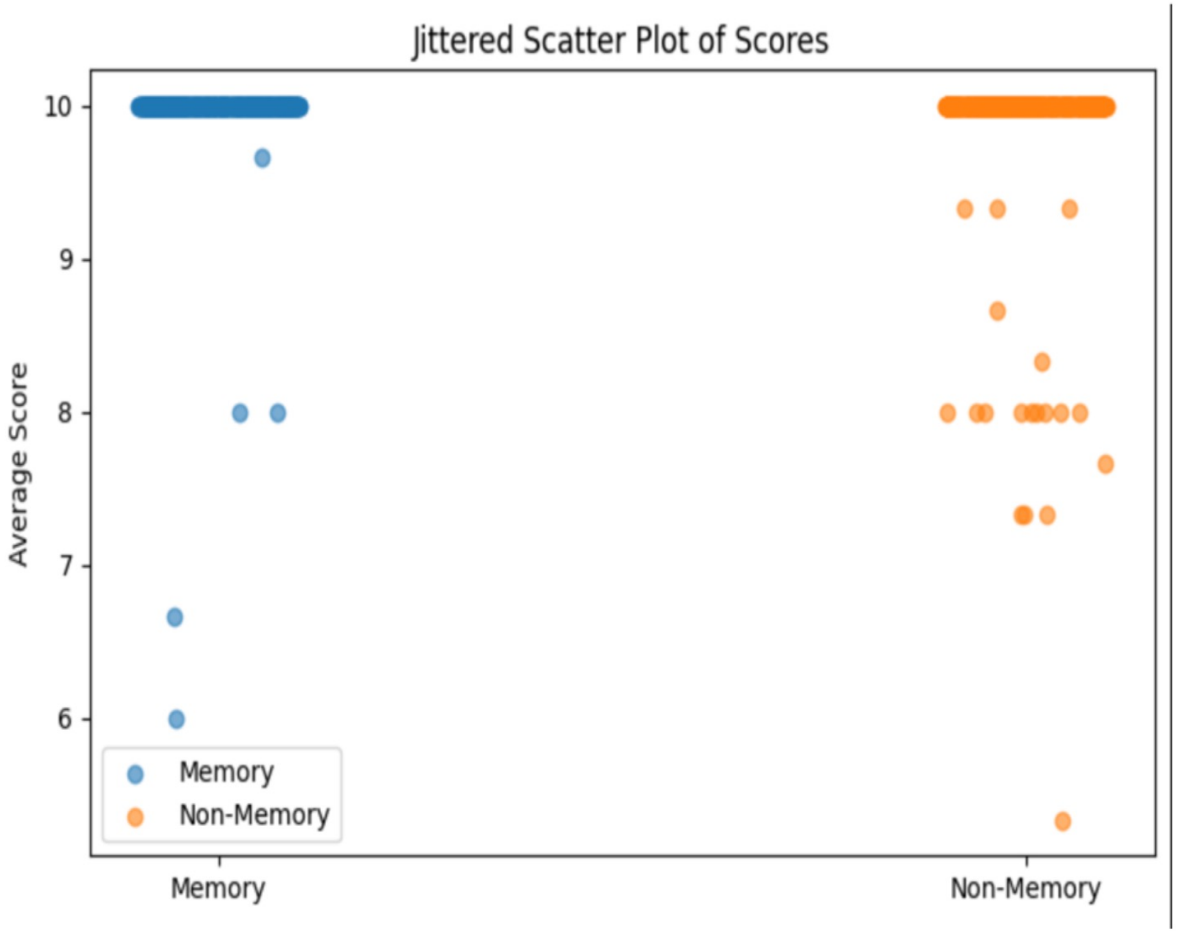


Fig. 1. Jittered scatter plot of scores comparing memory and non-memory models. Generated by the author using Google Colab, 2025. [Image file].

Personalization Factor

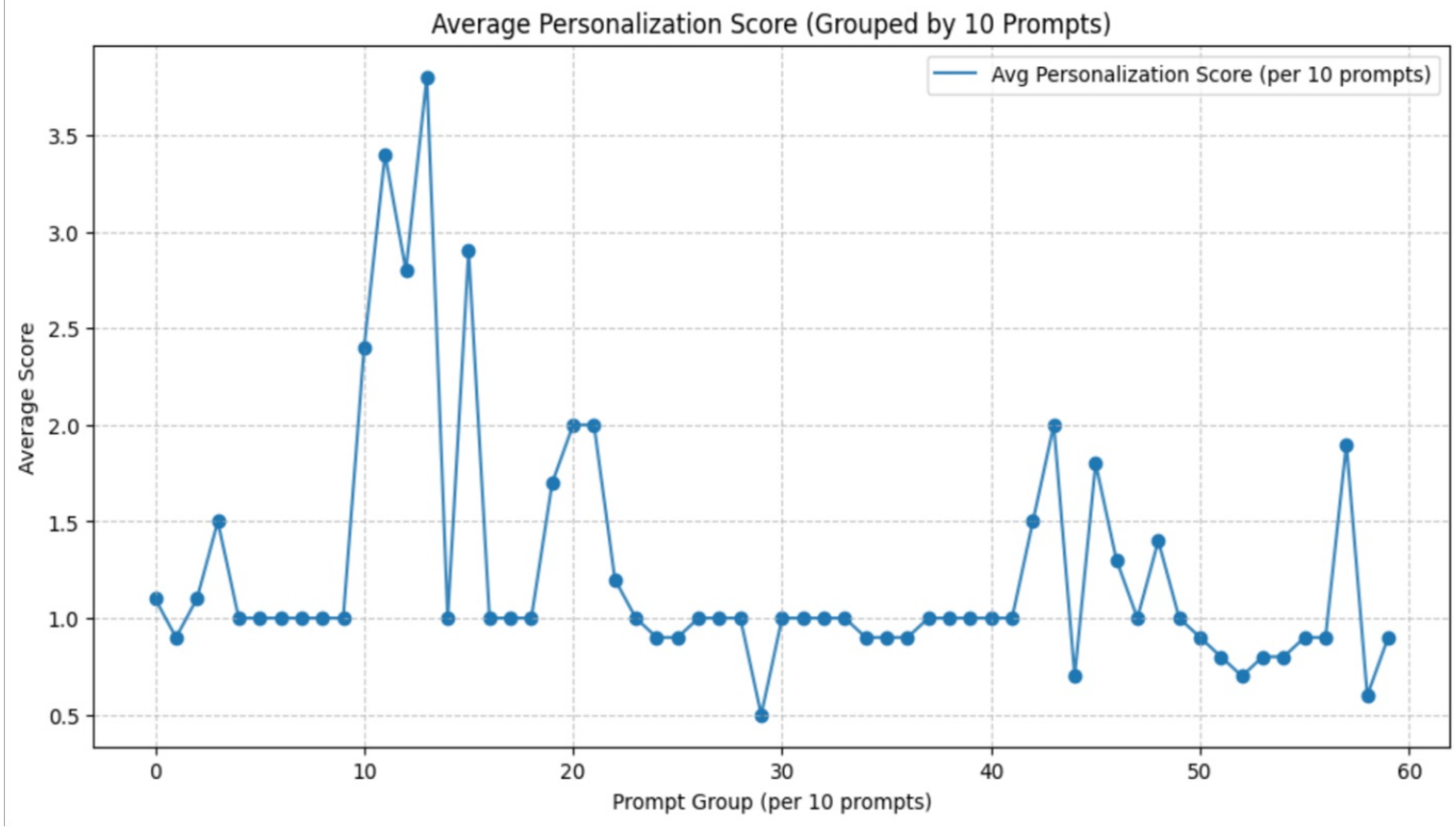


Fig. 3. Line plot showing average personalization scores grouped by 10 prompts. Generated by the author using Google Colab, 2025. [Image file].

Deciding

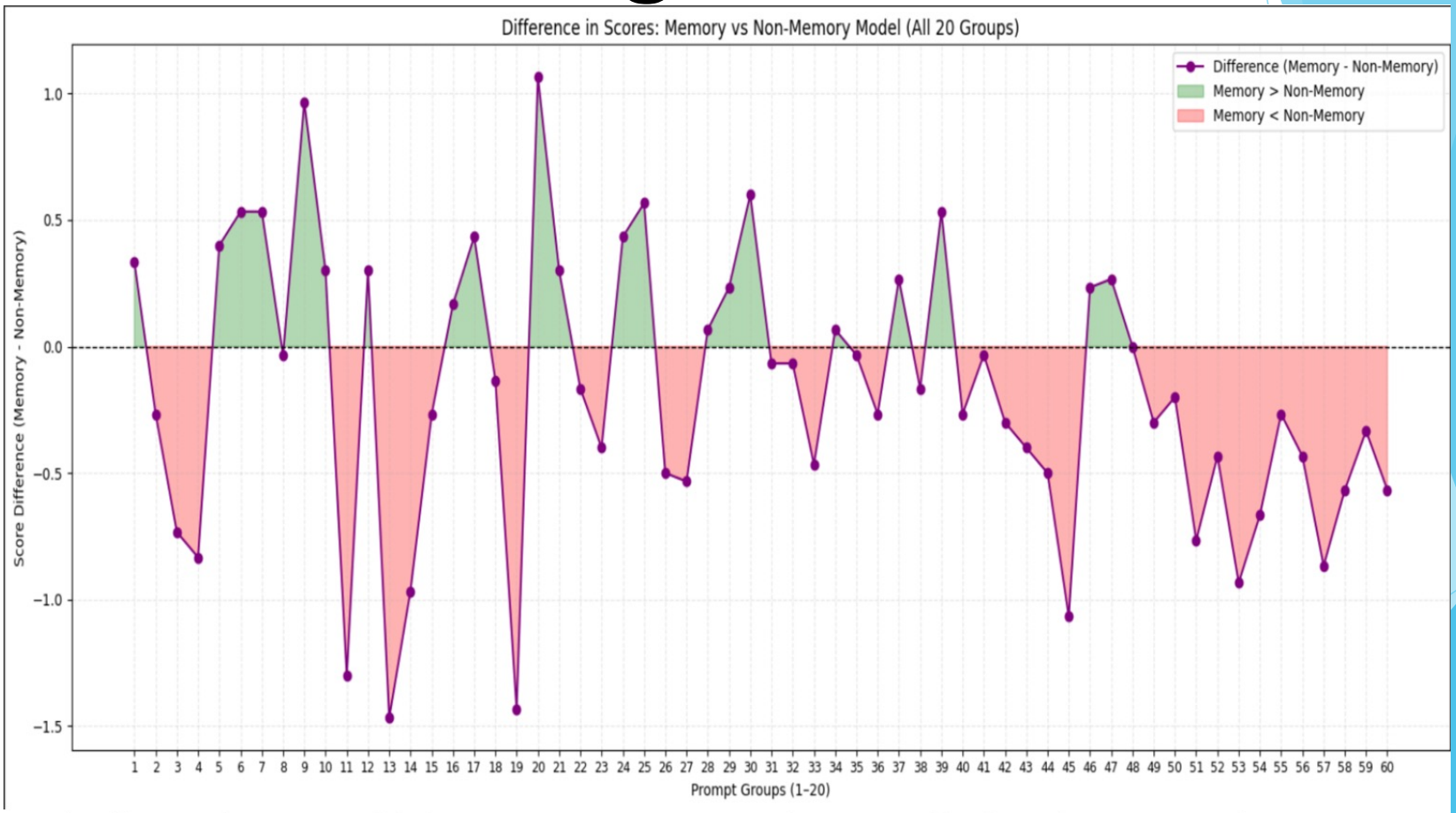


Fig. 5. Differences between Models (Positive = Non-Memory Better). Generated by the author using Google Colab, 2025. [Image file].

Deciding

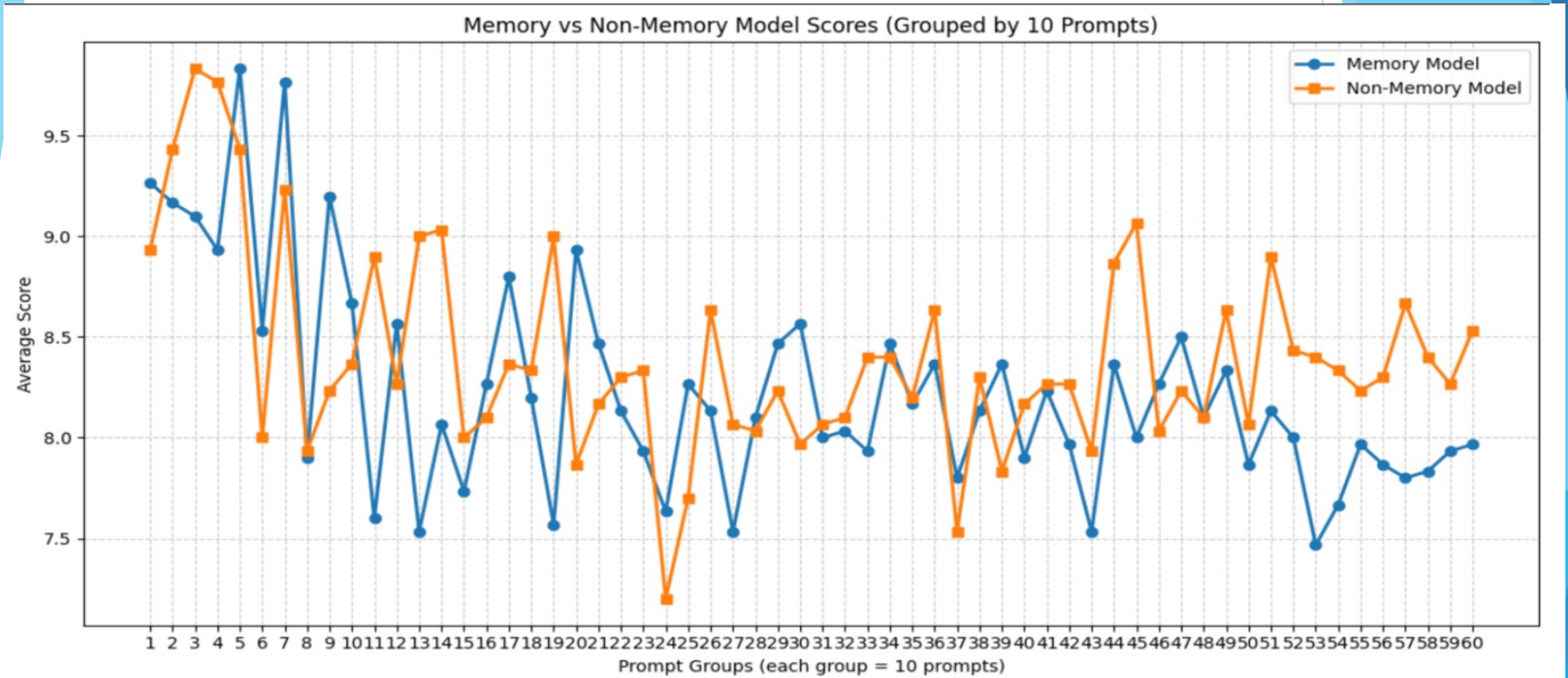


Fig. 4. Memory vs Non-Memory Model Scores (Grouped by 10 Prompts). Generated by the author using Google Colab, 2025. [Image file].

Personalization Factor

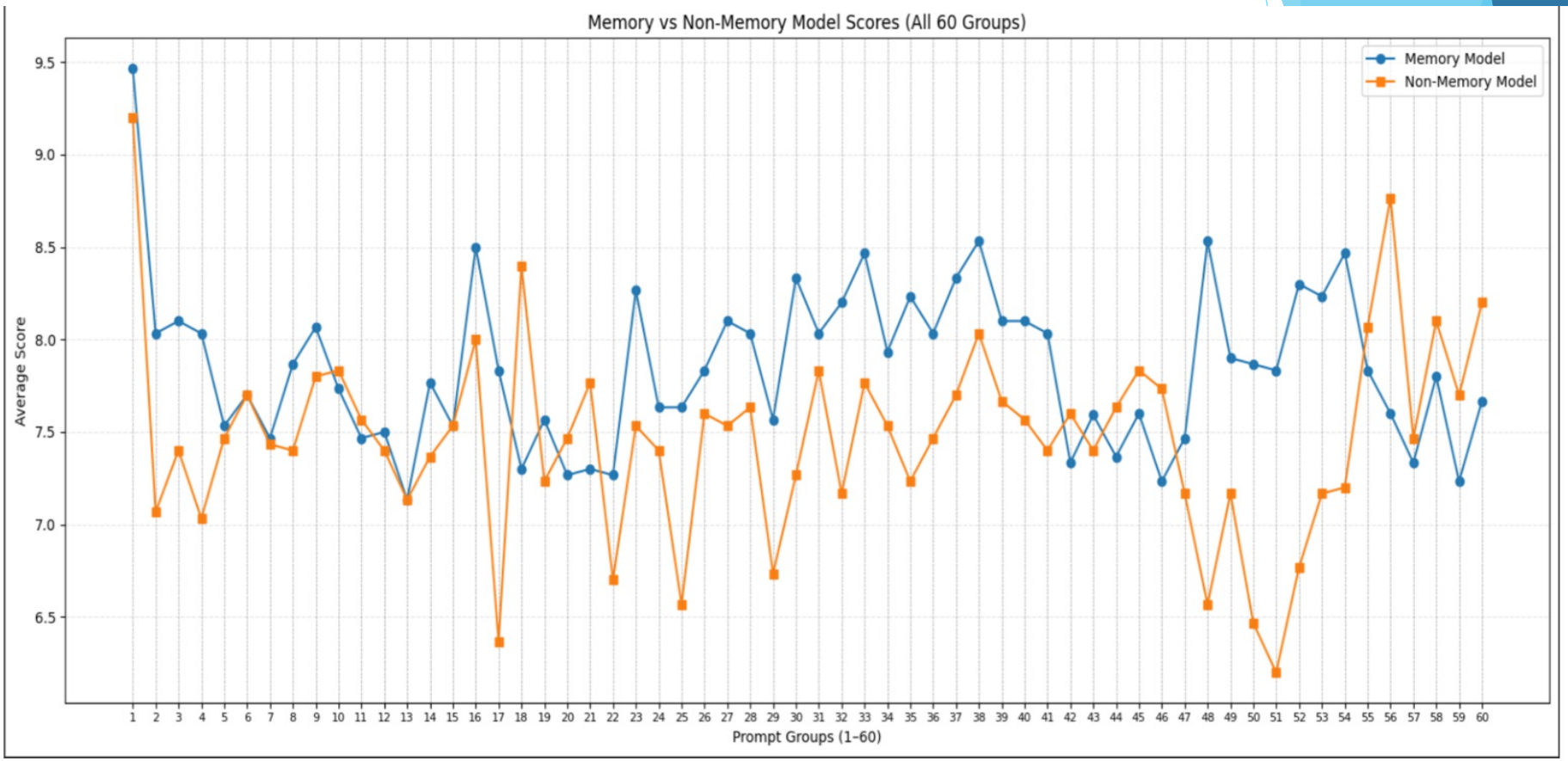


Fig. 7. Memory vs. Non-Memory Model Scores (All 60 Groups). Generated by the author using Google Colab, 2025. [Image file].

Correlation

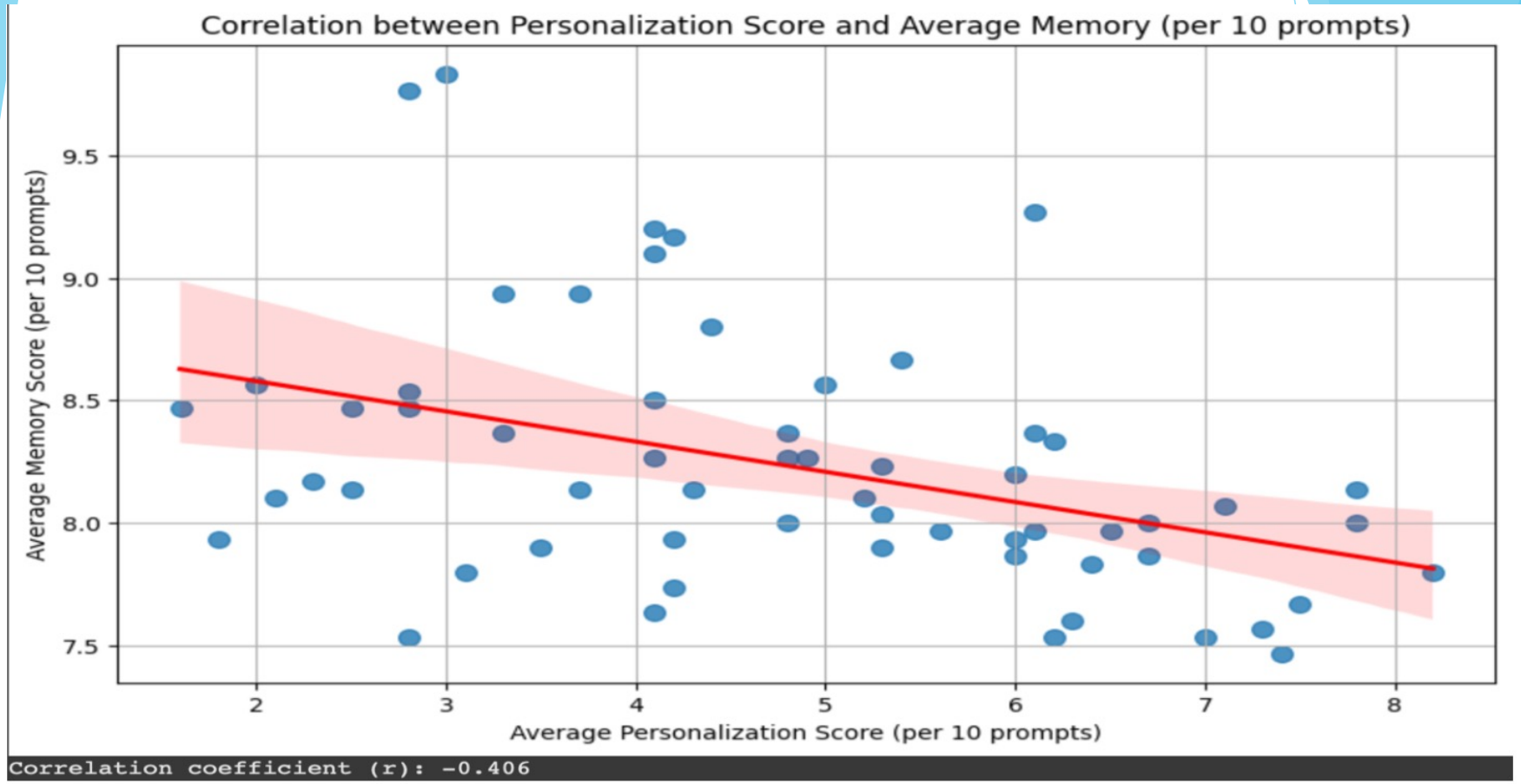


Fig. 6. Correlation between Personalization Score and Average Memory. Generated by the author using Google Colab, 2025. [Image file].

Describing

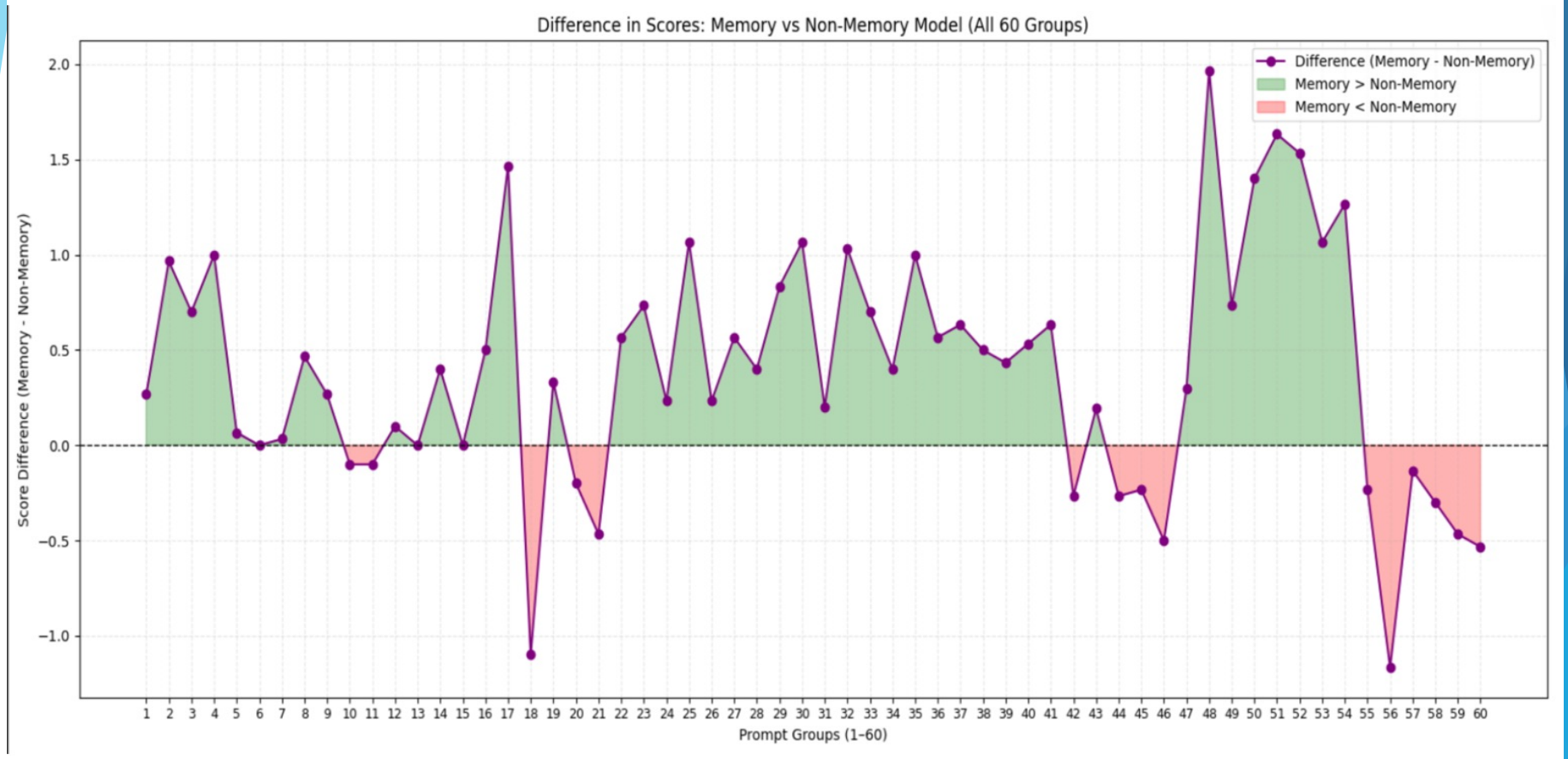


Fig. 8. Difference in Scores: Memory vs. Non-Memory Model (All 60 Groups). Generated by the author using Google Colab, 2025. [Image file].

Personalization

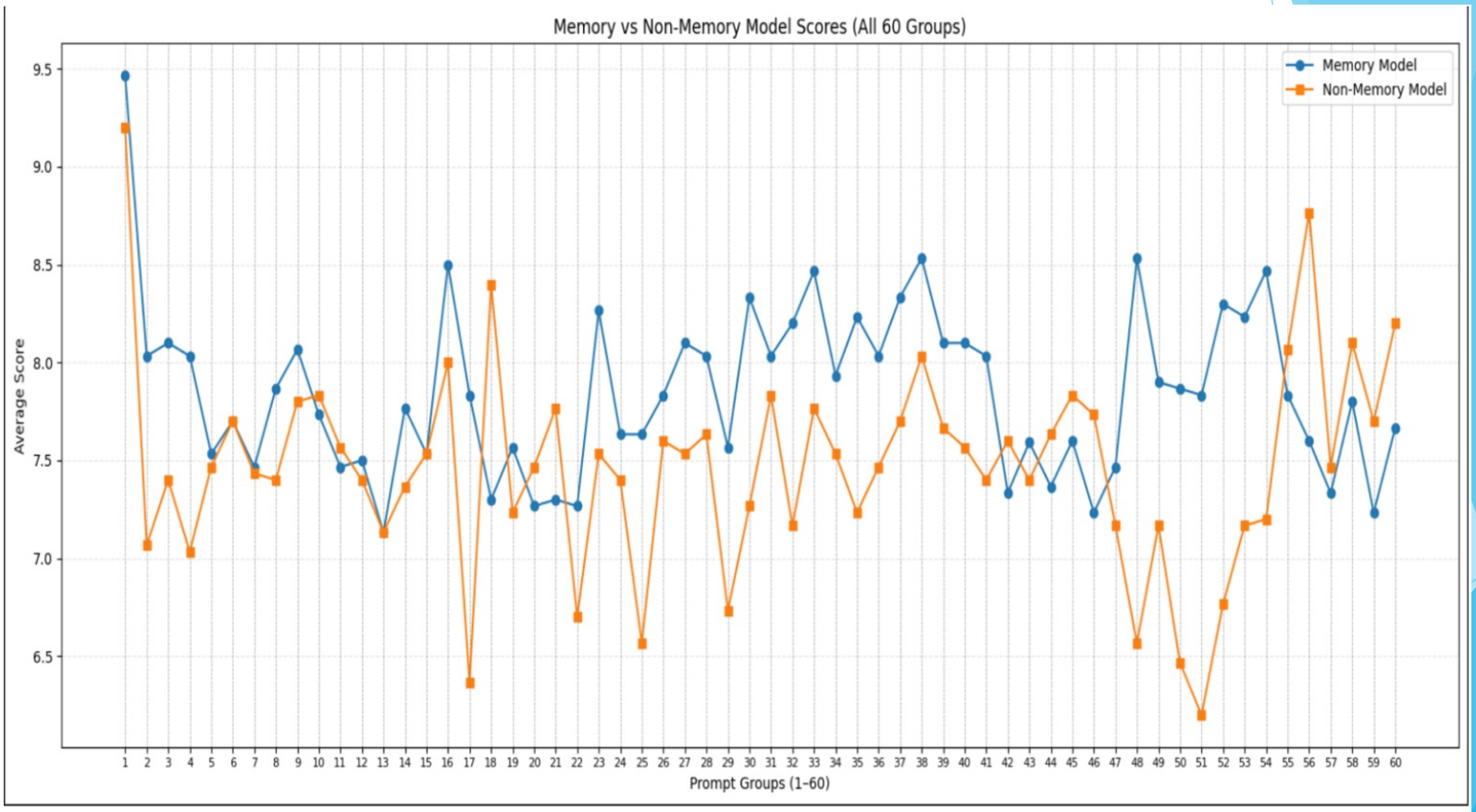


Fig. 7. Memory vs. Non-Memory Model Scores (All 60 Groups). Generated by the author using Google Colab, 2025. [Image file].

Correlation

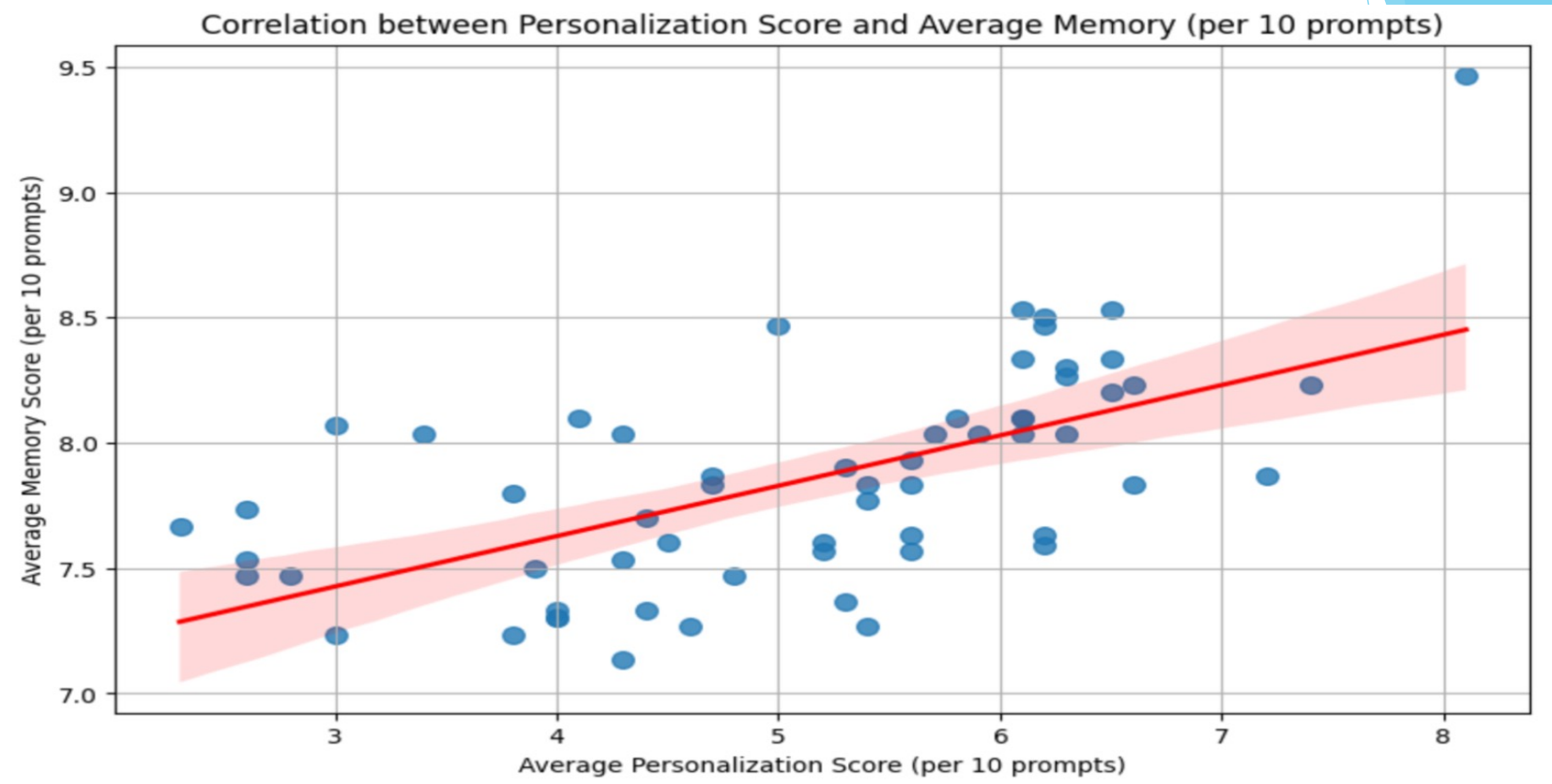


Fig. 9. Correlation between Personalization Score and Average Memory (r = 0.612). Generated by the author using Google Colab, 2025. [Image file].

Essay and Writing

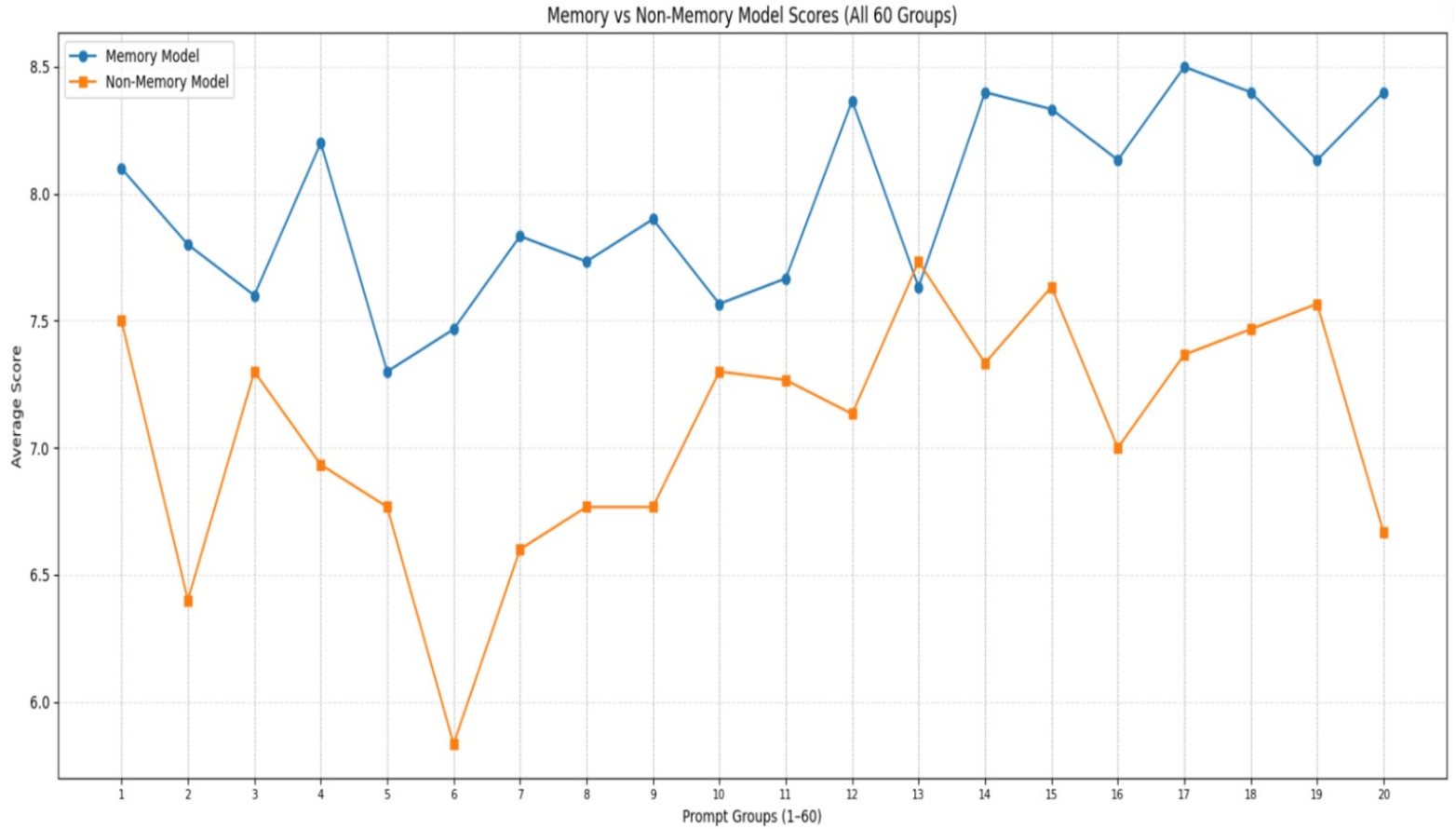
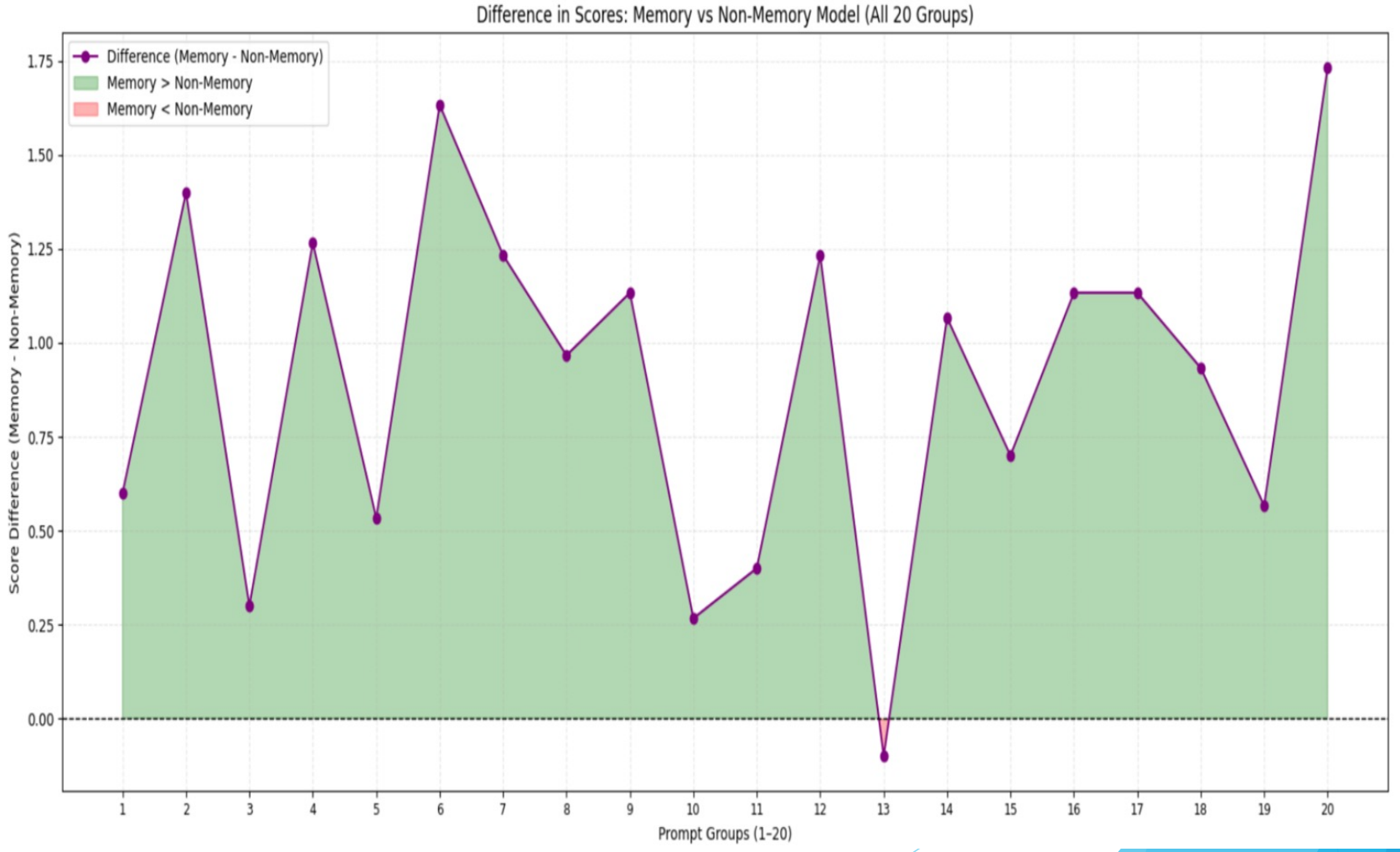


Fig. 10. Memory vs. Non-Memory Model Scores (First 20 Groups). Generated by the author using Google Colab, 2025. [Image file].

Essay and Writing



Essay and Writing

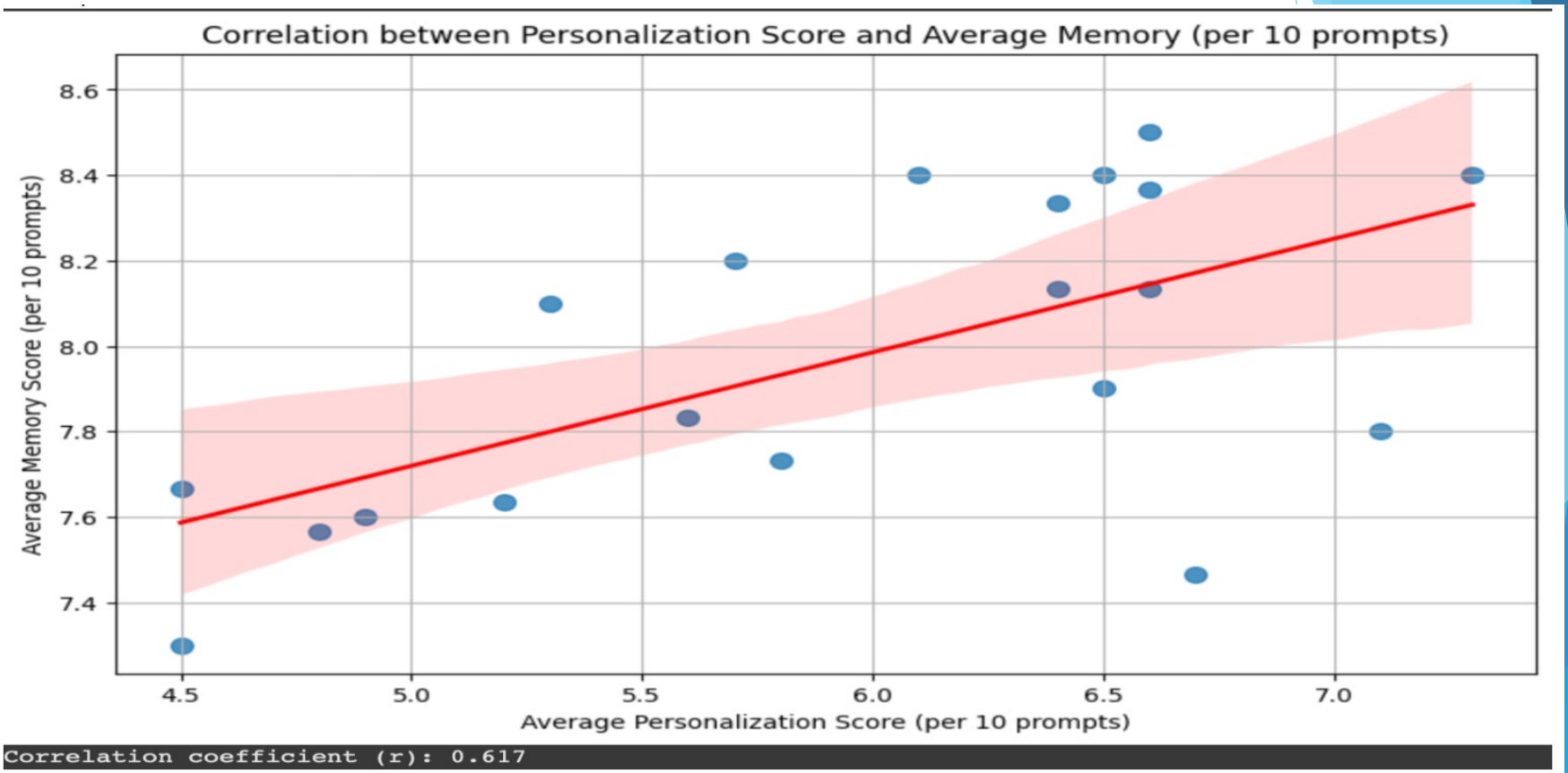


Fig. 12. Correlation between Personalization Score and Average Memory (r = 0.617). Generated by the author using Google Colab, 2025. [Image file].

Interpretation/Insights

- Memory does not consistently improve performance

- Non-memory models perform comparably in many tasks

Memory is most beneficial for:
- Descriptive tasks
- Writing tasks

In some cases:
- Memory introduces bias
- Assumes prior knowledge

Limitations

- Evaluation limited to only two models (ChatGPT, Google Gemini)
- Personalization factor includes subjective judgment
- Small dataset: only 2,000 prompts

Results may vary across:

- Different models
- Different user contexts

Thank You / Q&A

Questions?

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